

Assignment - 7 : File System & Path Module in Node.js

Objective

Create a simple Node.js program that reads a student file and displays information about the file using the **fs** and **path** modules.

Tasks -

1. **Create a new project folder -**
 - Create a folder named **NodeAssignment**.
 - Open this folder in VS Code.

2. **Create **app.js** -**
 - Inside **NodeAssignment** folder, create a file named: app.js

3. **Create a **data** folder -**
 - Inside **NodeAssignment**, create a folder named: data

4. **Create a student file -**
 - Inside the **data** folder, create: student.txt

5. **Add student information -**
 - Write the following information inside **student.txt**:

Name: John

Age: 21

Course: Node.js

6. **Import the required modules -**
 - In **app.js**, import:
 - File System (**fs**) module
 - Path (**path**) module

7. Create the file path -

- Use `__dirname` and `path.join()` to create the path of `student.txt`.

8. Read the file -

- Use `fs.readFile()` to read `student.txt`.
- Use `"utf8"` so that the file contents are displayed as readable text.

9. Handle errors -

- Check whether an error occurred while reading the file.
- If there is an error, display the error message.
- Stop the function using `return`.

10. Display student details -

- If the file is successfully read, display:

Student Details:

Name: Rahul

Age: 21

Course: Node.js

11. Display the file name -

- Use `path.basename()` to display:

12. File Name: student.txt

13. Display the directory -

- Use `path.dirname()` to display the directory where `student.txt` is located.

14. Display the file extension -

- Use `path.extname()` to display:

15. File Extension: .txt

Expected Output

Student Details:

Name: Rahul

Age: 21

Course: Node.js

File Name: student.txt

Directory: [your data folder path]

File Extension: .txt